Bureau
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Le système international d'unités
The International System of Units

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Annexe 2
Appendix 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.7
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Ytterbium 171 Ion ($f \approx 642\,\mathrm{THz}$)

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May 2016
mai 2016

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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second Part 2.7: Ytterbium 171 Ion ($f \approx 642\ \mathrm{THz}$)

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^{171}Yb ion, $^{2}S_{1/2} - {}^{2}F_{7/2}$ unperturbed optical transition

1. Recommended value [1] of the frequency

$f\left({}^{171}\mathrm{Yb}^{+},\ \mathrm{octupole}\right) = 642\,121\,496\,772\,645.0\ \mathrm{Hz}$

equivalent to

$\lambda\left({}^{171}\mathrm{Yb}^{+}\right) = 466\,878\,090.060\,497\ \mathrm{fm}$

with an estimated relative standard uncertainty of 6×10^{-16}.

This radiation is already endorsed by the CIPM as a secondary representation of the definition of the second.

2. Source data

Adopted value $f(^{171}\mathrm{Yb}^{+},\ \mathrm{octupole}) = 642\,121\,496\,772\,645.0\ \mathrm{Hz}$

$u_c/y = 6 \times 10^{-16}$

calculated from

$f(^{171}\mathrm{Yb}^{+})$ / Hz	u / Hz	$f(^{171}\mathrm{Yb}^{+}\ \mathrm{oct})\ /\ f(^{171}\mathrm{Yb}^{+}\ \mathrm{quad})$	u	source data
642 121 496 772 657.0	12			[2]
642 121 496 772 645.15	0.52			[3]
642 121 496 772 646.22	0.67			[4]
642 121 496 772 644.91	0.37			[5]
642 121 496 772 645.36	0.25			[6]
		0.932 829 404 530 964 65	3.1×10^{-16}	[5]

as a weighted mean from the above values applying the method of ref. [7]. Considering that the values came from only two laboratories, the CCTF considered it prudent to enlarge the uncertainty by a factor of two.

References

[1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies http://www.bipm.org/jsp/en/CIPMRecommendations.jsp

[2] K. Hosaka, S. A. Webster, A. Stannard, B. R. Walton, H. S. Margolis, and P. Gill: Frequency measurement of the $^2S_{1/2}$ -$^2F_{7/2}$ electric octupole transition in a single $^{171}Yb^+$ ion, *Phys. Rev. A* **79**, 033403 (2009).

[3] N. Huntemann, M. Okhapkin, B. Lipphardt, S. Weyers, Chr. Tamm, and E. Peik: High-Accuracy Optical Clock Based on the Octupole Transition in $^{171}Yb^+$, *Phys. Rev. Lett.* **108**, 090801 (2012).

[4] S. A. King, R. M. Godun, S. A. Webster, H. S. Margolis, L. A. M. Johnson, K. Szymaniec, P. E. G. Baird, and P Gill: Absolute frequency measurement of the $^2S_{1/2} - {}^2F_{7/2}$ electric octupole transition in a single ion of $^{171}Yb^+$ with 10^{-15} fractional uncertainty, *New J. Phys.* **14**, 013045 (2012).

[5] Godun, R. M.; Nisbet-Jones, P. B. R.; Jones, J. M.; King, S. A.; Johnson, L. A. M.; Margolis, H. S.; Szymaniec, K.; Lea, S. N.; Bongs, K. & Gill, P.: Frequency ratio of two optical clock transitions in $^{171}Yb^+$ and constraints on the time-variation of fundamental constants, *Phys. Rev. Lett.,* **113**, 210801 (2014).

[6] N. Huntemann, B. Lipphardt, Chr. Tamm, V. Gerginov, S. Weyers, and E. Peik: Improved Limit on a Temporal Variation of m_p/m_e from Comparisons of Yb^+ and Cs Atomic Clocks, *Phys. Rev. Lett.* **113**, 210802 (2014).

[7] H. S. Margolis, P. Gill: Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values; *Metrologia*, 2015, **52**, 628-634